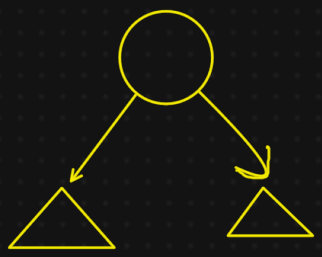
Construct a tree from inorder and preorder



inorder : left subtree | root | right subtree

4 2 5 | 1 | 3 6

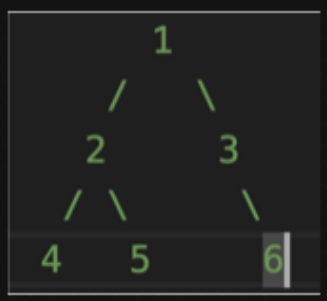
len

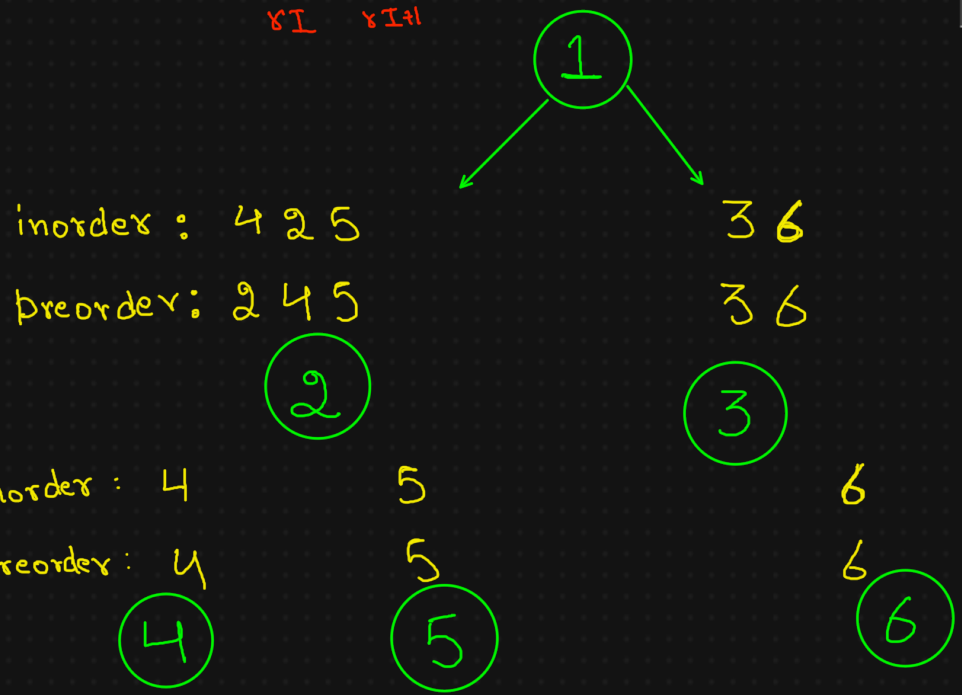
preorder: root | left subtree | right subtree

1 | 2 4 5 | 3 6

rI | rI+1

```
Preorder = 1 2 4 5 3 6
Post Order = 4 5 2 6 3 1
Inorder Traversal = 4 2 5 1 3 6
```





inS inE index

inorder : left subtree | root | right subtree

4 2 5 | 1 | 3 6

inrI

preorder: root | left subtree | right subtree

1 | 2 4 5 | 3 6

rI

preS

preE

inrI → inorder root index

Construct tree (inorder , preorder , inS , inE, prS , prE)

left lin S = inS lin E = inrI - 1 lpr S = preS + 1 lprE

right r InS = inrI + 1 rin E = inE rpr S = lprE + 1 rprE = prE

lprE - lprS = linE - linS rprE - rprS = rInE - rinS